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ARTICLE

Vote buying and negative agenda control: A problem for the study of money in politics

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Abstract

Politicians, pundits, and ordinary citizens frequently argue that money has an outsized influence on US politics. Yet research on the effect of money on politicians' behavior finds limited effects, especially on voting behavior. I construct a formal model to show that these limited effects may be an artifact of the institutional powers of agenda setters, and the strategic nature of the data generating process. The model predicts that, under general conditions, vote buying against proposals can occur with no vote held, and that vote buying is most likely when uncertainty is high, the interest group and agenda setter are extreme, and the pivotal legislator is moderate. I argue that these conditions held during the 2021 negotiations over the Build Back Better Act. Using a synthetic control design, I test whether the swing voters in the Senate received more campaign contributions than they would have otherwise, finding evidence consistent with the model.

On September 27, 2021, the House Committee on the Budget published its report on the Build Back Better Act (BBB)—a budget reconciliation package that contained many of President Biden's top priorities. With that report, BBB was officially on the congressional agenda. It was reported early on that BBB would have an easy time passing the House, as Democrats held a small but comfortable majority. It passed on November 19, 2021, with 220 voting for, and 213 voting against.[1]

There were, however, two Democratic holdouts in the Senate: Joe Manchin of West Virginia and Kyrsten Sinema of Arizona. With a 50–50 Democrat-Republican split in the Senate, BBB would not pass if either did not support it. The question of whether these two Senators would support the bill was a major news story for months (Beckett and O'Keefe, 2021; Fox et al., 2021). Sinema and Manchin also drew intense scrutiny of their campaign finances, as many news outlets insinuated that corporate interests were attempting to influence their votes on BBB (Schwartz, 2021; Reed, 2021). On December 19, 2021, Joe Manchin issued a statement saying that he would not support BBB in its current form. The bill effectively stayed dormant until August 2022, when Democratic leaders made a number of policy concessions, and BBB transformed into the Inflation Reduction Act, which passed through Congress and was signed into law in August 2022.

An important feature of this saga is that BBB never came to the floor for a vote in the Senate. Manchin's statement assured Democratic leaders that the vote would fail if it came to the floor, so they tabled the bill. This means that even if Sinema and Manchin's votes were influenced by special interests, we would not see those votes in the roll call data. This poses a challenge for the study of money in politics, as it would seem, from the researcher's perspective, that Sinema and Manchin received contributions for nothing in return. If successful vote buying in favor of the status quo occurs frequently, with agenda setters tabling such votes, researchers may be underestimating the true effect of campaign contributions on voting behavior due to missing data.

Verification Materials: The materials required to verify the computational reproducibility of the results, procedures, and analyses in this article are available on the *American Journal of Political Science* Dataverse within the Harvard Dataverse Network, at: https://doi.org/10.7910/DVN/DQABDX.

The Cornell Center for Social Sciences verified that the data and replication code submitted to the AJPS Dataverse replicates the numerical results reported in the main text of this article.

[1] https://clerk.house.gov/Votes/2021385

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This may, in part, explain the mixed empirical findings on whether campaign contributions influence the voting behavior of members of Congress. For example, Ansolabehere et al. (2003) review dozens of articles with the majority either finding null effects or effects in the opposite direction, that is, more money being associated with voting less favorably for the interest group. They also conduct their own analysis and find that contributions from business or labor political action committees (PACs) do not have a statistically significant effect on House members' propensity to vote in line with business interests. Other studies, in contrast, have found statistically significant effects in the expected direction (Dwyer and Stratmann, 2023; Grier et al., 2023; Stratmann, 2002). What all of the latter studies have in common is that they confine their analysis to a single piece of legislation on which multiple votes were held. The benefit of these more limited analyses from the perspective of the theoretical results here is that there is no missing data problem.

In order to better understand the conditions under which these roll call data may be missing, I construct a formal model of vote buying with agenda control. There are three players in the model: an agenda setter, a legislator, and an interest group. The agenda setter first proposes a policy to the legislator, whose preferences are private information, so that the agenda setter and the interest group are uncertain about the outcome of the vote at the proposal stage. After the proposal stage, the interest group can offer transfers to the legislator to influence their vote. Finally, the legislator decides how they plan to vote and informs the agenda setter, who then decides whether to bring the proposal to the floor for a vote at some cost. It is important to note from the outset that even though the model is set up as an explicit exchange between the interest group and the legislator, this is not meant to be an exact replication of the influence process. The model is meant to provide a simple way to study the strategic incentives of an interest group who is able to contribute resources to a legislator, and how these incentives interact with agenda setting power.

The model offers two important predictions for the empirical study of vote buying in legislatures: (i) Under general conditions vote buying against proposals can occur with no vote held and (ii) vote buying is most likely when uncertainty is high, the interest group and agenda setter are relatively extreme, and the legislator is relatively moderate. I argue that these conditions were likely satisfied in the BBB agenda setting process, which implies that Senators Manchin and Sinema ought to have received more campaign contributions after the introduction of BBB than they would have otherwise. While news articles suggested this may be the case, I test this claim empirically with a synthetic control design, finding that Sinema, but not Manchin received significantly more contributions as a result of BBB.

RELATED LITERATURE

The empirical question of whether campaign contributions affect legislators' votes has been studied extensively, but the results are largely inconclusive. Some studies suggest that effects are minimal (Ansolabehere et al., 2003), while other studies find significant effects (Dwyer and Stratmann, 2023; Grier et al., 2023; Stratmann, 2002). The purpose of the model presented here is to offer insights about the conditions under which successful vote buying ought to occur, especially when the interest group lobbies against the proposal and in favor of the status quo. In fact, one of the main conclusions drawn from the extensive work of Baumgartner et al. (2009) is that the most predictive feature of lobbying success is being on the side of the status quo. Importantly, in this paper, I show that these are exactly the scenarios in which such votes are unlikely to show up in the roll call data.[2] This would lead us to systematically underestimate the effect of campaign contributions on legislators' voting behavior due to the researcher observing transfers, but not observing votes that would have failed.

This paper contributes to a large literature on formal models of money in politics, and specifically to the literature on exchange models of lobbying.[3] Many models of lobbying are constructed to offer predictions about *who* is lobbied, rather than the conditions under which successful lobbying occurs. For example, Denzau and Munger (1986) predict that legislators whose constituencies are indifferent over the motivating policy of the interest group will be targeted more. Snyder (1991) studies one of the first spatial models of vote buying, and predicts that when a single vote buyer is present, they will target legislators near the median of the chamber. Groseclose and Snyder (1996) and Banks (2000) show that with competitive vote buyers moving sequentially, it can be optimal for the first mover to buy a supermajority of votes, so that it is not possible for the second mover to peel off a majority for themselves.

Other prominent models of lobbying focus on influence over a unitary actor (Che and Gale, 1998; Grossman and Helpman, 1996; Hillman and Riley, 1989). The most closely related models to mine are those where interest groups use transfers to buy access. Most of these models focus on an informational mechanism, whereby lobbyists buy access to then persuade a policymaker (Austen-Smith, 1995,

[2] The claim that agenda control can influence voting data is not new (e.g., Bateman et al., 2017; Clinton, 2012b, 2020). Yet, the consideration of the way in which vote buying interacts with agenda control to influence the data we observe is, I believe, novel.

[3] For a recent review, see Schnakenberg and Turner (2023).

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1998; Cotton, 2012; Lohmann, 1995). A notable exception is Judd (2023), who studies a legislative bargaining process where an interest group can probabilistically gain access to a proposer. If the group gets access, then they can offer the proposer a transfer in exchange for proposing a policy. Judd (2023) finds that the mere possibility of access changes the set of policies that could gain majority support in the committee.

Another closely related model is Blumenthal (2026), in which an interest group can expend effort to raise or lower the probability that a proposal succeeds. They find that the threat of spending to defeat a proposal can influence the proposer, leading to the threat not being carried out in equilibrium. Similar mechanisms are discussed in Dal Bó and Di Tella (2003), and Chamon and Kaplan (2013). As noted by Blumenthal (2026), this equilibrium behavior can explain Tullock's puzzle about why contributions to politicians are so low when the stakes industries have in policy are so high (Ansolabehere et al., 2003; Tullock, 1972). In these models, a politician makes a unilateral policy choice or campaign promise, and an interest group can contribute directly to the politician, or their opponent. I show that this unfulfilled threat mechanism is also at play in a model of agenda setting, where the proposer, who is not subject to direct interest group influence, must get the approval of a legislator, who can receive transfers from a group. My model adds to this body of work by specifying conditions under which successful vote buying is most likely and allowing for the agenda setter to exercise negative agenda control.

The current model also contributes to the formal theory of agenda setting and agenda control (Chiou and Rothenberg, 2003; Cox and McCubbins, 2005, 2007; Krehbiel, 2010; Romer and Rosenthal, 1978). In fact, the complete information version of the model here can be thought of as a simplified combination of Snyder (1991) and Romer and Rosenthal (1978). With incomplete information, the model allows us to consider how vote buying interacts with negative agenda control, that is keeping bills from coming to the floor. Gailmard and Jenkins (2007) find that majority parties in both the House and the Senate exercise negative agenda control to a similar and substantial degree. And Jenkins and Monroe (2012) and Jenkins and Monroe (2016) both mention vote buying as a cause of negative agenda control, though neither develop a general formal model.[4] It is the interaction of negative agenda control and vote buying that leads to the distorted inference about the effect of money on voting behavior.

[4] Jenkins and Monroe (2012) employ some formal-theoretic analysis to generate their hypothesis that party leaders will issue side-payments to moderate members of their coalition, compensating them for tabling bills they would have preferred to pass. But their analysis does not involve an outside group or a fully endogenous agenda.

A FORMAL MODEL OF AGENDA CONTROL AND VOTE BUYING

The discussion in the introduction suggests that there may be a problem for the empirical study of vote buying in legislatures, that is, that vote buying may occur, but not be fully observed in the data because of strategic control over the legislative agenda. In this section, I construct a formal model of agenda control and vote buying to determine when we should expect to observe vote buying in the data if it is occurring.

The model has three players: an agenda setter, S, an interest group, G, and a unitary actor legislative body, which I call the legislator or, L. There is a one-dimensional policy space, $X = [-1, 1]$, with a status quo policy given by $q \in X$. Each of the players has preferences over policy represented by a negative quadratic utility function:

$$u_i(x) = -(x - \hat{x}_i)^2,$$

where $\hat{x}_i$ is i's ideal policy. At times it will be useful to condense notation and use $\Delta_i(x)$ to represent the utility difference for i from a policy x from the status quo:

$$\Delta_i(x) = u_i(x) - u_i(q).$$

Costs and transfers enter into the players' payoffs linearly, that is, given a policy $x \in X$ if the legislator receives a transfer t from the group, then the legislator and group's payoffs, respectively, are:

$$u_L(x) + t \qquad \text{and} \qquad u_G(x) - t.$$

The simplest way to think of the transfer is as a monetary payment, but this could be any benefit the group can provide to the legislator. The complete payoff for the agenda setter will be described in more detail below.

I will start by analyzing a complete information baseline model, where all of the aspects of the game are common knowledge, and all previous actions are observed. In the subsequent incomplete information game, the only modification will be the addition of uncertainty over the location of L's ideal point. In particular, $\hat{x}_L$ will be drawn from a uniform distribution over the interval $[\mu_L - \epsilon, \mu_L + \epsilon]$, and only the legislator knows their true ideal point. The complete information model is the limiting case as ϵ goes to zero. Thus, when there is uncertainty, the agenda setter and the group are not sure exactly which policies the legislator will support, and the size of the transfer needed for them to vote in line with the interest group.

The agenda setter moves first by choosing a policy proposal, $p \in X$. Second, the group decides which

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policy to support, $z \in \{p, q\}$, and can make a contingent transfer offer, $t \in \mathbb{R}_+$ to the legislator. Third, the legislator decides whether to vote for or against the proposal ($d = 1$ or $d = 0$ respectively), considering the policy proposal p, the policy the group supports, z, and the transfer offer t. Fourth, the agenda setter observes whether the legislator decided to vote for or against the proposal, and chooses whether to bring p to the floor for a vote ($b = 1$) or table the proposal ($b = 0$). If the agenda setter brings the proposal up for a vote and it fails, they pay a cost $c > 0$. One can think of the proposer learning the legislator's decision through a whipping process, or through public statements made by the legislator, as happened with BBB.[5] Finally, p is implemented if S decides to bring it to the floor for a vote and it passes, otherwise q remains the status quo. Note that I am suppressing the formal vote of the legislator as it is equivalent to their decision. The interest group pays the transfer if and only if the legislator *decides* to vote for their preferred policy, z. The agenda setter's payoff at the end of the game is:

$$U_S(p, z, t, d, b) = \begin{cases} d \cdot u_S(p) + (1 - d) \cdot (u_S(q) - c) & \text{if } b = 1 \\ u_S(q) & \text{if } b = 0. \end{cases}$$

Importantly, in an equilibrium, the agenda setter may decide not to hold a vote because the legislator decides to vote against the proposal. I am assuming that the group still provides the transfer if the group's preferred outcome is that the proposal fails. That is, the transfer is decision-contingent rather than vote-contingent. If the transfer were, instead, vote-contingent, and the agenda setter knew how the legislator would vote, then the interest group could never influence the legislator to vote *against* a proposal. Vote buying against a proposal occurs when the group makes an offer to persuade the legislator to vote against a policy they prefer. But if no vote is held, then there is no way for the group to compensate the legislator for their policy loss, so the legislator would prefer to vote for the policy.

To summarize, the timing of the complete information game is as follows:

1. S proposes $p \in X$.
2. G chooses a policy to support $z \in \{p, q\}$, and offers a transfer $t \in \mathbb{R}_+$.
3. L decides whether to vote for or against p ($d = 1$ or $d = 0$, respectively).
4. S chooses whether or not to bring the proposal up for a formal vote ($b = 1$ or $b = 0$, respectively).
5. The status quo prevails if and only if either no vote is held or the L votes against the proposal. Payoffs are realized.

In the incomplete information game there is a Period 0, in which Nature draws the legislator's ideal point, $\hat{x}_L$, from the uniform distribution on $[\mu_L - \epsilon, \mu_L + \epsilon]$, and only the legislator observes the realized value.

A strategy for S is a pair, (ρ, β), where ρ is the policy choice in the first period, and $\beta(p, z, t, d)$ is a function that maps the policy choice by S, the policy support and transfer from G, and L's decision of whether to vote for the proposal or not, into a choice of whether to bring the bill to the floor or not. In the complete information analysis, I will suppress the final choice of whether to bring the proposal to the floor, as it is redundant. L's decision is similarly a function, $\delta(\hat{x}_L, p, z, t)$, that depends on the proposal, and the group's actions, but L can also condition their behavior on Nature's draw of $\hat{x}_L$. Finally, G's strategy is a pair, (ζ, τ), where $\zeta(p)$ is a function mapping proposals into a support decision, and $\tau(p)$ is a function mapping proposals into transfers. A strategy profile will be denoted $\sigma = (\rho, \zeta, \tau, \delta, \beta)$.[6]

The case of misaligned interests

In order to focus attention on the empirically relevant predictions of the model, I restrict the main analysis to a particular ordering of ideal points and the status quo. I say that there is "group-legislature misalignment" when the legislator and the agenda setter are to the left of the status quo, and the group is to the right of the status quo. I assume that the agenda setter is more extreme than the legislator, reflecting the idea that the agenda setter is likely a party leader elected by the party, whereas the median legislator of the legislature will be in the majority party, but more moderate than a party leader. When there is uncertainty over the ideal point of L, I assume that the configuration of ideal points and status quo holds with probability 1. While the interest group and the agenda setter do not know the value of $\hat{x}_L$, the location of all other parameters and the distribution of $\hat{x}_L$ are common knowledge.

Definition 1. There is **group-legislature misalignment** if ideal points and the status quo are ordered such that:

$$\hat{x}_S < \hat{x}_L < q < \hat{x}_G.$$

If the status quo were between the agenda setter's ideal point and the legislator's ideal point, there would

[5] I will discuss potential alternatives to this set-up in the comments on the model.

[6] I provide a formal description of the conditions for a subgame perfect Nash equilibrium (SPNE) in Online Appendix A (p. 2).

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be policy gridlock regardless of whether the group is in the model. And when all of the players' interests are aligned in, say, moving the status quo to the right, then negative agenda control does not occur. Thus, the case of group-legislature misalignment is the relevant one to consider to demonstrate how negative agenda control interacts with vote buying. I will formally introduce and briefly discuss the case of aligned interests later on. The full analysis of this case is located in Online Appendix C (pp. 12–15).

Comments on the model

There are a number of strong assumptions in the model that deserve discussion up front. First, I assume that there is only a single interest group in an effort to simplify the empirical predictions of the model. It is, nevertheless, an important question of how those predictions might change when more than one vote buyer is present. If there are many vote buyers on the same side of a policy issue, then a free-riding incentive may come into effect. And if there are vote buyers on opposite sides of the policy issue, then the choice of how to model the vote buying process becomes important, whether it is an all-pay contest, or first price auction, and so forth. Furthermore, there is, empirically, a large imbalance between the amount that business interest groups contribute, and the contributions from any other type of group. There is, of course, heterogeneity within the designation "business interest groups," but the current model is focused on the asymmetric nature of the interest groups' preferences, and I am abstracting away from any free-rider problems.

Second, I assume that there is only a single legislator. This reflects the goal of this model, which is to study the extent of vote buying in an institution with agenda control, and not to characterize the features of legislators who might be targeted by vote buying. By simplifying the model to include only a single legislator, we can sharpen focus on the most relevant empirical predictions.

Third, I assume that the agenda setter perfectly observes the legislator's decision to vote for or against the proposal. In the complete information case, this does nothing at all, as S already knows what policies L prefers. In the incomplete information case, this assumption is meant to highlight the discrepancy in knowledge of the agenda setter at the policy proposal stage, and the stage when they decide whether to bring the proposal up for a vote. A consequence of this stark assumption is that no vote should ever fail. This is, of course, empirically false, but one could simply add additional uncertainty at the agenda control stage, whereby the agenda setter only observes a noisy signal of the legislator's intention. Yet, that added realism would not change the relevant empirical predictions of the model. There are other scenarios in which an agenda setter may want to hold a vote—although they know it will fail—for example, to force members to cast unpopular votes, or to use the failed vote as a signaling device. While these incentives are interesting, they are beyond the scope of the model.

An alternative assumption would be that the legislator delivers a cheap talk message to the agenda setter indicating how they intend to vote. Suppose also that the group's transfer is contingent on the message they deliver to the agenda setter. Under this set of assumptions, there would exist an equilibrium in which the legislator truthfully reveals their intention. To see this, note that if given the policy, the transfer offer, and the realized value of the legislator's ideal point, the legislator prefers to vote for the proposal, then it is strictly better for them to tell the agenda setter the truth. If instead they prefer to vote against the proposal (which will only happen because they are compensated by a transfer), then they prefer to tell the agenda setter that they will vote against the proposal so that they indeed receive the transfer that makes voting against the proposal optimal. Therefore, telling the truth would be an equilibrium.

COMPLETE INFORMATION BENCHMARK

Despite the fact that negative agenda control does not occur in the complete information model due to the fact that the agenda setter can perfectly foresee what the legislator will do, it provides intuition for the incomplete information model. Moreover, a number of interesting results arise even in this simplified setting. In particular, I find that there is no vote buying in equilibrium, but policy is nevertheless biased toward the group.

As usual, we can solve for the SPNE using backward induction. Since the agenda setter's ideal point is to the left of the status quo, any proposal $p > q$ is weakly dominated for S.[7] I will therefore only focus on proposals, $p \leq q$, that are not weakly dominated for S.

The legislator decides to vote for the policy proposed by the agenda setter if and only if the utility the legislator receives from the proposal is strictly greater than the utility of the status quo plus the transfer offer from the group, that is,

$$u_L(p) > u_L(q) + t.$$

The inequality is strict here because the group always prefers the status quo to any undominated proposal, and in order for the group to have a well-defined best

[7] These proposals are weakly dominated in the strategic form equivalent of the extensive form game.

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response, the legislator accepts the group's offer (votes against the proposal) when indifferent.

Lemma 1 provides the optimal strategy of G in the subgame following the policy proposal. Intuitively, it says that G will always prefer the status quo to the policy proposed by S, and if a strictly positive transfer is made, then it will exactly compensate L for voting against a policy that they prefer. The condition $\Delta_L(p) < -\Delta_G(p)$ says that the size of the transfer cannot be larger than the group's net loss from the proposal relative to the status quo. This inequality is strict because, in equilibrium, the group does not provide a transfer when indifferent, ensuring the agenda setter has a proper best response.

Lemma 1. *Given an undominated proposal p, the optimal strategy for G is given by $\zeta^*(p) = q$, and*

$$\tau^*(p) = \begin{cases} \Delta_L(p) & \text{if } \Delta_L(p) < -\Delta_G(p) \text{ and } \Delta_L(p) \geq 0 \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

Proof. All proofs are located in Online Appendix B (pp. 2–12). □

Lemma 2 characterizes the condition under which the group finds it profitable to buy votes. If the midpoint of the proposal and the status quo is strictly less than the midpoint of G and L's ideal points, then it is strictly profitable for G to buy L's vote against the proposal.

Lemma 2. *For an undominated proposal, $p < q$, the inequality $\Delta_L(p) < -\Delta_g(p)$ holds if and only if:*

$$\frac{p+q}{2} < \frac{\hat{x}_L + \hat{x}_G}{2}.$$

It is possible that there is no undominated policy that S can propose that will pass. That is, every policy $p < q$ will elicit vote buying from G. In Condition 1, I maintain that S proposes the status quo in this case.

Condition 1. If there is no proposal that passes with positive probability and is not weakly dominated, then S proposes q.

There are a number of reasons why this is an appropriate equilibrium selection condition. First, it could be microfounded by adding a small cost to proposing a policy. Second, it simplifies the analysis of the model, as the policy proposal is always the policy outcome in the complete information benchmark. Third, it is the only condition that ensures continuity of the equilibrium policy choice as a function of the exogenous parameters. Finally, it stacks the deck against equilibrium vote buying. If, for example, I maintained that S chooses their ideal point in this case, then there could be vote buying, but the proposal would never pass.

Proposition 1 characterizes the unique SPNE satisfying Condition 1. The value $\hat{p}$ is the point at which G is indifferent between buying votes and not. In a sense, we can think of G and L as an aggregate agent with ideal point $\hat{x}_{GL} = \frac{\hat{x}_L + \hat{x}_G}{2}$, and S as proposing a policy that is accepted or rejected by the aggregate agent. Note that $\hat{p} = 2\hat{x}_{GL} - q$, that is, the point at which the aggregate agent is indifferent between accepting and rejecting the proposal. As $\hat{x}_S < \hat{p}$, it is well-known that S will select $\hat{p}$ when proposing to the aggregate agent. Therefore, $\hat{p}$ is the equilibrium policy proposal under group-legislature misalignment provided that $\hat{p} < q$, otherwise it is weakly dominated, and by Condition 1, S proposes q.

Proposition 1. *Let $\hat{p} \equiv \hat{x}_G + \hat{x}_L - q$. There is a unique pure strategy SPNE, σ^*, satisfying Condition 1. The optimal policy proposal is: $\rho^* = \min\{\hat{p}, q\}$. G's strategy is given in Equation (1) in Lemma 1. There is no vote buying in equilibrium. L votes for ρ^* if and only if $\Delta_L(\rho^*) > \tau^*(\rho^*)$.*

Despite there being no vote buying in equilibrium, policy outcomes are biased toward the interest group, relative to the model without the interest group. Corollary 1 gives the expression for the amount of bias introduced by the interest group's threat of buying votes. The first term of the expression in Corollary 1, $\min\{\hat{p}, q\}$, is the equilibrium policy given by Proposition 1, whereas the second term, $\max\{2\hat{x}_L - q, \hat{x}_S\}$, is the equilibrium policy from the game without the interest group.

Corollary 1. *The policy bias introduced due to G's ability to buy votes is*

$$\min\{\hat{p}, q\} - \max\{2\hat{x}_L - q, \hat{x}_S\} > 0.$$

In their proposal, S needs to move policy far enough to the right to make G indifferent between buying votes and not, and this is always farther to the right than they would have to go to make L indifferent between accepting and rejecting the proposal. Figure 1 depicts the policy bias characterized in Corollary 1 for two configurations of ideal points. The left two points in both cases are the two possible equilibrium policies in the model without the group, and the two points between L and G's ideal points are the two possible equilibrium policies in the model with the group.

This dynamic of agenda setters biasing policy to reduce equilibrium vote buying may explain, to some extent, why it appears that transfers to politicians are much lower than we would expect, given the large

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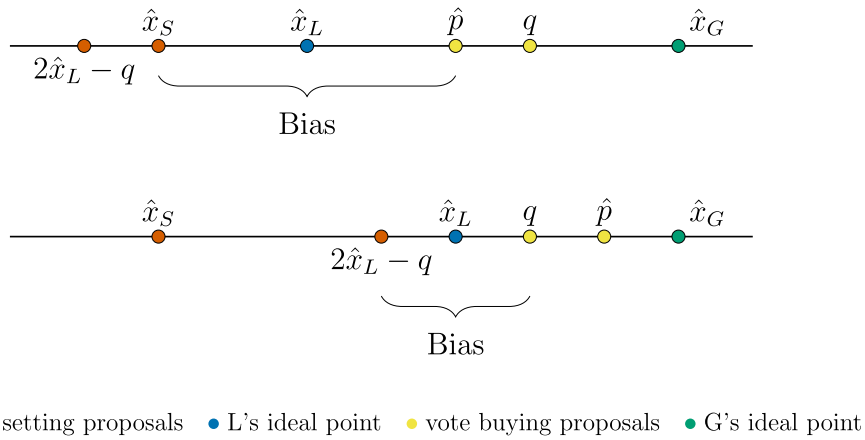

FIGURE 1 Policy bias with the interest group. *Notes*: This figure depicts the policy bias induced by the group's ability to buy votes.

stakes interest groups have in policy (Ansolabehere et al., 2003; Tullock, 1972). Agenda setters take into account potential opposition by interest groups, and choose policy to balance their own ideological objectives with the likelihood of their proposal passing. Thus, by the time voting takes place, the opposition is quieted. In the complete information limit, no transfers occur at all because the group is indifferent between buying votes and not. However, as I will show in the next section, transfers can occur in equilibrium when there is uncertainty over the legislator's preferences.

Furthermore, the policy bias here matches with the empirical findings of Clinton (2012a), who shows that canonical agenda setting models do not fit the data when considering policy changes to the minimum wage. In particular, he finds that the gridlock interval is larger than canonical models predict, and that conditional on policy change occurring, the changes are smaller than they predict. As most proposed changes to the minimum wage are proposals to raise it, there are well-funded interests who prefer the status quo over those proposals. Corollary 1 then suggests that the gridlock interval will be larger than in the canonical agenda setting model, and conditional on a policy passing, it will be biased toward the status quo.

UNCERTAINTY OVER THE LEGISLATOR'S PREFERENCES

The stark prediction that no equilibrium transfers should occur is somewhat artificial due to the assumption of complete information in the previous section, where the agenda setter knows exactly how to set policy to perfectly balance the incentives of the legislator and the interest group. Yet, the story of BBB, and most other important legislation, is one of uncertainty. In particular, the interest group and the agenda setter may be uncertain about the exact nature of the legislator's policy preferences. In such a case, the agenda setter may be willing to risk some chance that their proposal is killed by the group, for the prospect of getting a better policy. To capture the uncertainty over the legislator's preferences, I let $\hat{x}_L$ be a random variable with a uniform distribution over the support $[\mu_L - \epsilon, \mu_L + \epsilon]$, with $\epsilon > 0$. The main substantive difference between the findings of the complete and incomplete information models is that in the latter, there can now be vote buying in equilibrium, in which case the agenda setter does not hold a formal vote.

The decision of whether to bring the proposal up for a vote is mechanical, since S observes L's decision. Again, this could be through a public announcement like Joe Manchin made about BBB, or it could be through a whipping process, in which the legislator credibly conveys their vote intention. Therefore, S brings an undominated proposal to the floor if and only if $d = 1$. From L's perspective, their decision of how to vote remains unchanged from the complete information benchmark, but from S and G's point of view, L's decision is now stochastic. In order to condense notation, let $\pi(p, t)$ be the equilibrium probability that L decides to vote for the policy, p, given an offer, t.[8]

As before, G will always prefer the status quo to any proposal that is not weakly dominated for S. In choosing the optimal transfer offer, G maximizes their expected utility, that is, they solve

$$\max_{t \in \mathbb{R}_+} \pi(p, t) u_G(p) + (1 - \pi(p, t))[u_G(q) - t]. \tag{2}$$

Lemma 3 characterizes G's optimal transfer as a function of the proposed policy. Just as in the complete

[8] Formally, $\pi(p, t) \equiv \Pr[\delta^*(\hat{x}_L, p, z, t) = 1]$, where δ^* is a sequentially rational decision for L and we are integrating over $\hat{x}_L$. Under group-legislature misalignment, note that $z = q$.

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information model, there will still be a region of policies such that G can offer a transfer to guarantee that L decides to vote against those policies. There will also still be a region of policies that are relatively close to the status quo for which the amount that G would have to spend to lower the probability that L votes for those policies exceeds G's willingness to pay. Now, with uncertainty, there is a middle region of policies such that G's optimal offer results in L deciding to vote for policies in that region with a strictly interior probability. I denote the lower and upper bound of this region of policies by $\underline{p}$ and $\overline{p}$, respectively.

More specifically, note that a policy $p \leq 2(\mu_L - \epsilon) - q$ does not occur in equilibrium since this is the case when L is voting against p with probability 1 in the absence of any transfer from G. And when $p \geq \overline{p}$, the value of t that maximizes G's expected utility still leads to L voting for the policy, so they do not find it profitable to provide a transfer.[9] The transfer $\mathbb{E}[\Delta_L(p)] + 2\epsilon(q-p)$ is a threshold value for which any transfer weakly above that value *guarantees* L will vote against the proposal. G finds it optimal to provide this transfer for any proposal sufficiently far to the left, that is, $p < \underline{p}$. Finally, the offer $(q-p)[\hat{x}_G - (\mu_L + \epsilon)]$ is the transfer that maximizes G's expected utility when $\pi(p,t) \in (0,1)$.[10]

Lemma 3. *Define* $\hat{p}_\mu \equiv \hat{x}_G + \mu_L - q$, *and* $\underline{p} \equiv \hat{p}_\mu - 3\epsilon$, *and let* $\overline{p} \equiv \hat{p}_\mu + \epsilon$. *Given an undominated proposal, p, the optimal strategy for G is given by* $\zeta^*(p) = q$ *and*

$$\tau^*(p) = \begin{cases} 0 & \text{if } p \leq 2(\mu_L - \epsilon) - q \text{ or } \overline{p} \leq p \\ \mathbb{E}\left[\Delta_L(p)\right] + 2\epsilon(q-p) & \text{if } 2(\mu_L - \epsilon) - q < p < \underline{p} \\ (q-p)\left[\hat{x}_G - (\mu_L + \epsilon)\right] & \text{if } \underline{p} \leq p < \overline{p}. \end{cases} \tag{3}$$

Figure 2 shows G's offers as a function of policy, $\tau^*(p)$, and it shows the induced probability that L will decide to vote for the policy given the optimal transfer, $\pi(p, \tau^*(p))$. One can see that the interval $(\underline{p}, \overline{p})$ is the set of policies such that the equilibrium probability, $\pi(p, \tau^*(p))$, is strictly between zero and one. I will therefore call $\underline{p}$ and $\overline{p}$ the left and right *corner points* of $\pi(p, \tau^*(p))$. These points and the function, $\pi(p, \tau^*(p))$, will be critical to determining the equilibrium proposal from the agenda setter.

When S is considering a proposal to put forth, they need to anticipate what offers G would make in

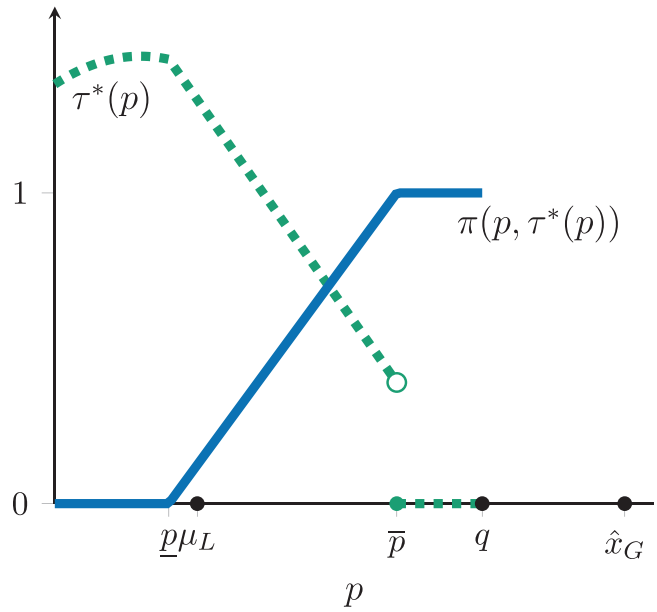

FIGURE 2 Offers and induced probability. *Notes*: This figure depicts the offers (solid blue) from the group as a function of policy and the induced probability (dashed green) of the legislator voting for the policy.

response, and how this policy and transfer influence L. Thus, S will take $\pi(p, \tau^*(p))$ as the given probability that a proposal p will succeed. They then choose a policy that maximizes their expected utility, that is, they solve the following problem:

$$\max_{p \in X} \pi(p, \tau^*(p)) u_S(p) + (1 - \pi(p, \tau^*(p))) u_S(q), \tag{4}$$

where $\tau^*(p)$ is given by Equation (3) in Lemma 3. Proposition 2 characterizes the unique SPNE of the game satisfying Condition 1.

Proposition 2. *There exists a unique pure strategy SPNE,* σ^*, *satisfying Condition 1. The equilibrium policy proposal is:*

$$\rho^* = \begin{cases} q & \text{if } \underline{p} \geq q \\ \tilde{p} & \text{if } \underline{p} < q \text{ and } \tilde{p} < \overline{p} \\ \overline{p} & \text{otherwise,} \end{cases}$$

where $\tilde{p}$ *is the interior solution the agenda setter's maximization problem (4). The vote buying strategy of G is given by Equation (3) in Lemma 3. There is strictly positive vote buying if and only if* $\rho^* = \tilde{p}$. *L votes for* ρ^* *if and only if* $\Delta_L(\rho^*) > \tau^*(\rho^*)$. *S brings* ρ^* *to the floor for a vote if and only if* $d^* = 1$.

In order to understand the equilibrium proposals in Proposition 2, it is helpful to think of S as having preferences over policy and the probability that L decides

[9] G is technically indifferent between offering any amount between zero and the value that maximizes their expected utility, since the transfer is never actually provided. I select the equilibrium in which G offers nothing.

[10] It is straightforward to see that as ϵ converges to zero, the diameter of the set $\{p : \underline{p} \leq p < \overline{p}\}$ goes to zero, $\mathbb{E}[\Delta_L(p)] + 2\epsilon(q-p)$ converges to $\Delta_L(p)$, and $\overline{p}$ and $\overline{p}$ converge to $\hat{p}$. Therefore, we recover the complete information baseline as ϵ goes to zero.

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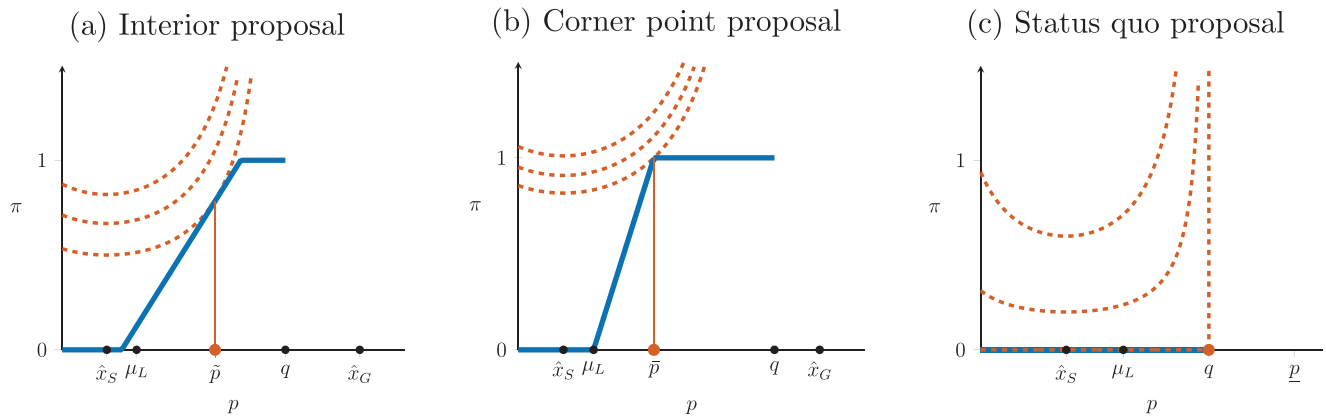

FIGURE 3 Agenda setter's maximization problem. *Notes*: The dashed orange curves represent indifference curves for the agenda setter over policy (x-axis) and probability of success (y-axis). The solid curves represent the probability of success as a function of policy.

to vote for their proposal. Of course, if S were completely unconstrained in setting the probability, they would choose their ideal point, $\hat{x}_S$, and choose the probability that L decides to vote for $\hat{x}_S$ to be one. However, S is constrained by the equilibrium behavior of G and L. They can only choose probability and policy pairs on the curve $\pi(p, \tau^*(p))$.

We can therefore visualize the agenda setter's choice as a constrained maximization problem where they choose both a policy, p, and a probability π, subject to the constraint that $\pi = \pi(p, \tau^*(p))$.[11] In Figure 3, the horizontal axis is the policy space, and the vertical axis is probability. The solid curve is $\pi = \pi(p, \tau^*(p))$, and the dashed lines are the agenda setter's indifference curves over policy and probability. That is, each indifference curve is the set of all policy and probability pairs that give the agenda setter the same level of expected utility.

I will call the point at which the agenda setter's indifference curve is tangent to the equilibrium probability, $\pi(p, \tau^*(p))$, the *interior proposal* or $\tilde{p}$. The full expression for this point as a function of exogenous parameters is given in the proof of Proposition 2 in Online Appendix B (p. 8). This case is depicted in panel (a) of Figure 3. Under some configurations of parameters, however, the best combination of policy and probability the agenda setter can achieve is at the point $\bar{p}$, which passes with probability 1. In this case, the indifference curves will not be tangent to the equilibrium probability, but will instead hit the right corner point. An example of this is shown in panel (b) of Figure 3. Finally, when $\underline{p} > q$, all undominated proposals have zero probability of passing. By Condition 1, S proposes the status quo in this case, which is depicted in panel (c) of Figure 3.

One of the primary goals of this model is to characterize when vote buying against the status quo occurs in equilibrium. To that end, I define the vote buying region as the set of status quo policies for which G makes a strictly positive offer to L to vote against the policy proposed by S. The only time strictly positive offers occur in equilibrium is when the optimal policy proposal is the interior proposal, $\tilde{p}$, which occurs when the status quo is sufficiently far to the right, and $\tilde{p}$ passes with a strictly interior probability. We must also make sure that the restrictions on the parameters for group-legislature misalignment are still satisfied. Specifically, we need $\mu_L + \epsilon < q$ and $q < \hat{x}_G$.

In Figure 4, I plot the equilibrium policy proposals and the probability that these proposals pass as functions of the status quo. The vote buying region can be seen as the middle region where the equilibrium proposal is increasing in the status quo, and it passes with a strictly interior probability.

Definition 2. The **vote buying region** is the set of status quo points for which G makes strictly positive offers in equilibrium under group-legislature misalignment. Formally,

$$\mathcal{V} = \left\{ q \in (\mu_L + \epsilon, \hat{x}_G) : \underline{p} < q, \tilde{p} < \bar{p} \right\}.$$

Proposition 3 shows that the set of status quo points satisfying these conditions is always a nonempty open interval. That is, unlike the case of complete information where there is never vote buying in equilibrium regardless of where the status quo lies, there always exists an open interval of status quo points such that strictly positive offers are made in equilibrium. Proposition 3 also provides some intuitive comparative statics on the length of this interval. In

[11] Formally, S solves

$$\max_{(p,\pi)} \pi u_S(p) + (1-\pi) u_S(q)$$
$$\text{s.t. } \pi = \pi(p, \tau^*(p)).$$

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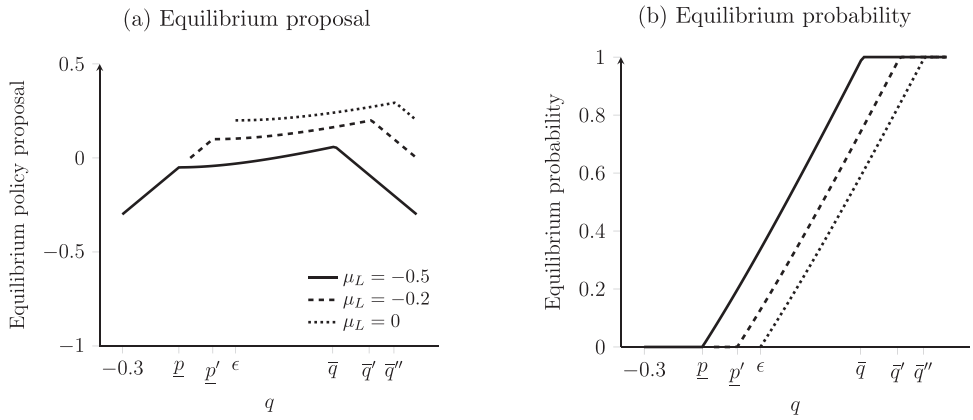

FIGURE 4 Equilibrium proposal and probability of passage as a function of the status quo. *Notes*: Panel (a) graphs the equilibrium proposal as a function of the status quo for three values of μ_L. Panel (b) graphs the equilibrium probability of success for the same values of μ_L.

particular, as G or S become more extreme, the length of the interval increases. However, if μ_L or ϵ become too large, they mechanically start to reduce the size of the vote buying region as they become even larger, due to the restriction that all status quo points must satisfy $q \in (\mu_L + \epsilon, \hat{x}_G)$ under group-legislature misalignment. So, the set of status quos for which there is vote buying in equilibrium decreases because the total possible set of status quos that satisfy group-legislature misalignment decreases. The condition, $\mu_L < \hat{x}_G - 5\epsilon$, ensures that the lower bound of the vote buying region is $\frac{\hat{x}_G + \mu_L - 3\epsilon}{2}$. Therefore, as long as ϵ and μ satisfy this inequality, the lower end point of the vote buying region does not run into the bounds imposed by group-legislature misalignment. And in this case, the length of the vote buying region is increasing in the expected location of L's ideal point, and the degree of uncertainty.

Proposition 3. *The vote buying region is the nonempty open interval*

$$\mathcal{V} = \left(\max\left\{ \frac{\hat{x}_G + \mu_L - 3\epsilon}{2}, \mu_L + \epsilon \right\}, \min\left\{ \overline{q}(\epsilon, \mu_L, \hat{x}_S, \hat{x}_G), \hat{x}_G \right\} \right),$$

where $\overline{q}$ is the unique point such that $\tilde{p}(\overline{q}) = \overline{p}(\overline{q})$. The length of this interval is increasing in $\hat{x}_G$, and decreasing in $\hat{x}_S$. If $\mu_L < \hat{x}_G - 5\epsilon$, then the length of the interval is increasing in μ_L and ϵ.

Qualitatively, Proposition 3 suggests that vote buying against policy proposals is most likely under the following circumstances: uncertainty is high, the legislator is relatively moderate, and the interest group and agenda setter are relatively extreme. These conditions seem to all be satisfied in a case like BBB, where Senators Manchin and Sinema were relatively moderate, the Democratic party leaders and organized business interests were relatively extreme, and the set of policies that both Manchin or Sinema would accept was highly uncertain.

Before turning to the empirical analysis, it is worth noting some findings and connections from the case of group-legislature alignment.

Definition 3. There is **group-legislature alignment** when:

$$q < \hat{x}_L < \hat{x}_S < \hat{x}_G.$$

In this case, all of the players in the game want to move the status quo to the right. With complete information, S is always able to obtain their ideal point. If L prefers S's ideal point to the status quo, then there is no vote buying. But if S's ideal point is too far to the right for L, then G will compensate L for the difference in utility from $\hat{x}_S$ and q. Thus, there can be vote buying in the aligned case, even in the complete information model.

When there is uncertainty over L's preferences, there are some cases in which S cannot achieve their ideal point. Nevertheless, they are able to get policy closer to their ideal point than they would in a model without the group. I define the vote buying region for this case in an analogous way, and find that the comparative statics on the length of the vote buying interval are nearly identical to the misaligned case.

EMPIRICAL ANALYSIS OF BUILD BACK BETTER ACT

The main empirical implication of the model is that vote buying is most likely when the legislator is moderate, the agenda setter is extreme, and there is substantial uncertainty over how the legislator will

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vote. And, unlike the complete information version of the model, vote buying can occur under group-legislature misalignment. I have argued that the negotiations over the BBB closely satisfies all of these conditions. There were months of news articles and television segments discussing whether BBB would pass in the Senate, Joe Biden was being pressured by progressive Democrats to "go big," and business interests were staunchly opposed to key provisions of the bill (Nichols, 2021).

Thus, the model predicts that at least one of Joe Manchin and Kyrsten Sinema ought to have received more campaign contributions than they would have if BBB had not been on the agenda. Note that it does not predict which one, since there is only a single legislator in the model. While news outlets reported increased contributions from corporate interests to Manchin and Sinema, they never made reference to the counterfactual level of contributions that we ought to have observed. I therefore test the claim that Manchin or Sinema received outsized campaign contributions due to BBB being on the agenda. Conventional methods of estimation such a difference-in-difference design or linear regression would be inappropriate here due to the small number of treated units. The synthetic control method (SCM), in contrast, allows us to estimate the counterfactual in exactly these cases (Abadie and Gardeazabal, 2003; Abadie et al., 2010, 2015).

In the current context, the SCM involves finding a weighted average of all other senators' monthly campaign contributions in order to estimate what Sinema or Manchin's campaign contributions would have been, had BBB not been on the agenda, or had they not been pivotal. Following the convention in the causal inference literature, I will refer to the introduction of BBB as the "treatment," and to Sinema and Manchin as the "treated units," as they were the pivotal legislators.[12] The pretreatment period is the 9 months of 2021 prior to the introduction of BBB, and the posttreatment period ends at the end of January 2022.[13]

In the SCM employed here, the weights are chosen to minimize the distance between the vector of monthly pretreatment contributions of the treated units and the weighted average of the pretreatment contributions of the untreated units.[14] The treatment effect is then the difference between the observed posttreatment contributions and the posttreatment contributions predicted by the weighted average of the untreated senators' contributions.

To simplify the analysis, I match only on the pretreatment values of the outcome variable and do not include controls. Furthermore, each pretreatment period is given equal weight in the analysis. This is termed "constrained regression," by Doudchenko and Imbens (2016), and is employed, for example, in the study of the value of political connections by Acemoglu et al. (2016). In the standard SCM, one can include predictors other than just the pretreatment outcomes, and there is a joint optimization process that chooses both weights on untreated units (also called "donor units") and weights on predictors.[15] Adding predictors to the model, however, comes at a significant computational cost (Kuosmanen et al., 2021), and decreases the transparency of the results (Ferman et al., 2020).

The difficulty is that there is no consensus on how to choose predictor weights, and Kuosmanen et al. (2021) show that the commonly used methods are numerically unstable. Moreover, in order to meaningfully include covariates, the researcher must omit some periods of pretreatment outcomes; otherwise the commonly used method for assigning predictor weights will assign zero weight to the covariates (Kaul et al., 2015). Ferman et al. (2020) argue that because there is no consensus on how to choose predictor weights, researchers have many opportunities for cherry-picking the covariates and the predictor weights that give them the desired result. They therefore "specifically recommend focusing on the specification that uses all the pre-treatment outcome lags as matching variables, unless there is a strong prior belief that it is crucial to balance on a specific set of covariates" (p. 513).

Another potential concern with the SCM is that of overfitting (Abadie and Vives-i Bastida, 2022). With a small number of pretreatment periods, and a large number of donor units, the SCM can achieve an extremely good fit in the pretreatment period, but the fit is spurious and leads to an estimation of large posttreatment effect even when none exists. There is, however, no exact guidance on how many pretreatment periods is enough, or how many donor units is too many. For context, in the analyses surveyed by Ferman et al. (2020), the median number of pretreatment periods is 9, and the median number of donor units is 37. I conduct a number of robustness checks to address concerns about overfitting and the lack of covariates located in Online Appendix D (pp. 16–21). Reassuringly, the main estimates are consistent across all specifications.

To formally introduce the method used here, the outcome variable of interest is the campaign contributions from PACs to senator j in period t, which I denote

[12] While an outline of BBB was agreed to earlier than September 27, 2021, the timeline of when or if the bill would make it to the congressional agenda was uncertain until then. I show that the synthetic control results for Sinema are robust to setting the treatment earlier in Online Appendix D (pp. 22–23).

[13] I extend the posttreatment period for 1 month after Manchin came out publicly against BBB to allow for contributions that may have come as reward.

[14] The weights are constrained to be nonnegative and sum to one in order to prevent extrapolation.

[15] I describe the standard SCM in more detail in Online Appendix D (pp. 19–21).

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c_{jt}. These data come from Open Secrets (OpenSecrets, 2025). The main analysis includes both direct contributions and independent expenditures supporting a candidate, but the main results are unchanged by dropping independent expenditures.

I bin the contributions by month to smooth out some of the volatility, and I examine those made throughout 2021, and include those from January 2022. Thus there are 13 time periods, $t \in \{1, \dots, 13\}$, and 100 units (senators), $j \in \{1, \dots, 100\}$. As BBB was introduced to the agenda on September 27, 2021, I set the treatment period to 10, so there are nine pretreatment periods and four posttreatment periods. I label Joe Manchin as unit 1 and Kyrsten Sinema as unit 2.

Let $\mathbf{c}_j^{pre}$ denote the 9×1 vector of pretreatment monthly campaign contributions for senator j, and let $\mathbf{c}_j^{post}$ be the 4×1 vector of posttreatment monthly contributions. Furthermore, let $\mathbf{C}_{-\{1,2\}}^{pre}$ be the 9×98 matrix of contributions in the pretreatment periods for all senators other than Manchin and Sinema, let $\mathbf{C}_{-\{1,2\}}^{post}$ be the similarly defined 4×98 matrix for the posttreatment periods, and let $\mathbf{C}_{-\{1,2\}}$ be the full 13×98 matrix of contributions for all periods.

The process of finding weights is then to find $\mathbf{w}_1^*$ and $\mathbf{w}_2^*$ that minimize the Euclidean distance between the pretreatment values of the treated unit and the convex combination of the pretreatment values, that is, for each $i = 1, 2$, $\mathbf{w}_i^*$ solves

$$\min_{\mathbf{w}_i \in \mathbb{R}^{98}} \quad \left\| \mathbf{c}_i^{pre} - \mathbf{C}_{-\{1,2\}}^{pre} \cdot \mathbf{w}_i \right\|$$

$$\text{subject to} \quad w_{i,j} \geq 0 \quad \text{for } j = 3, \dots, 100,$$

$$\text{and} \sum_{j=3}^{100} w_{i,j} = 1.$$

Once the weights are derived, the synthetic path of contributions for unit i is given by the dot product: $\mathbf{C}_{-\{1,2\}} \cdot \mathbf{w}_i^*$, where each entry of this vector is the estimated control contributions for period t, denoted $\hat{c}_{it}$.[16] Then the estimated treatment effect for $i = 1, 2$ is given by the sum of the differences between the actual contributions and the estimated counterfactual contributions over the posttreatment periods:

$$\hat{\tau}_i = \sum_{t=10}^{13} (c_{it} - \hat{c}_{it}).$$

Figure 5 plots the observed PAC contributions for Manchin (panel a) and Sinema (panel b) from January 2021 through January 2022, along with the synthetic PAC contributions for these two senators. The pretreatment fit for Manchin is much worse than it is for Sinema. Nevertheless, it appears that there was a small effect of BBB on Manchin's PAC contributions, but a relatively large effect for Sinema. Specifically, $\hat{\tau}_1 = \$55,935.61$ and $\hat{\tau}_2 = \$130,704.10$.

Statistical inference with the SCM is based on comparisons of the treatment effects with the effects estimated on placebo units. That is, we estimate the effect for all senators as if they were treated, and compare the effect sizes to the actual treatment effects.[17] One could, however, see a large treatment effect due to a poor pretreatment fit. To prevent making inferences from cases with poor pretreatment fit, we can adjust for the fit of the model, which is measured by the root mean squared error (RMSE) of the prediction of the synthetic relative to observed pretreatment contributions. I will refer to the pretreatment RMSE for j as $PreRMSE_j$, and the corresponding post prediction error as $PostRMSE_j$:

$$PreRMSE_j = \sqrt{\frac{\sum_{t=1}^{9} (c_{jt} - \hat{c}_{jt})^2}{9}}$$

$$PostRMSE_j = \sqrt{\frac{\sum_{t=10}^{13} (c_{jt} - \hat{c}_{jt})^2}{4}}.$$

There are two ways to adjust for pretreatment fit. The first is to truncate the placebo units to those with a pretreatment fit within some range of the treated unit. For example, in their most restrictive case, Abadie et al. (2010) truncate units with more than two times the PreRMSE. In the analysis here, that would only yield 6 placebo units for Sinema, but 82 for Manchin. So, in panel (b) of Figure 6, I plot the placebo units within 20 times the PreRMSE of Kyrsten Sinema. One can clearly see that Sinema has the largest treatment effect of any plotted placebo unit, with the exception of one observation having a higher value for the month of January. In contrast, panel (a) of Figure 6 shows the same analysis for Joe Manchin, except the placebo units plotted are only those with a PreRMSE less than Manchin's, which is still more than two thirds of all senators. Here, Manchin does not stand out among those with similar pretreatment fit, but Sinema's increase in contributions is large even relative to senators with substantially worse pretreatment fit.

The other way to adjust for pretreatment fit seen in the literature on synthetic controls (e.g., Abadie et al., 2015, 2010) is to compare the ratios of the PostRMSE to the PreRMSE, which I will refer to the root mean

[16] I solve the optimization problem using the *Rsolnp* package in *R*. This does significantly better than the *Synth* package, suggesting issues with the *Synth* package other than those identified by Kuosmanen et al. (2021).

[17] When estimating placebo effects, Manchin and Sinema are, of course, excluded from the donor pool.

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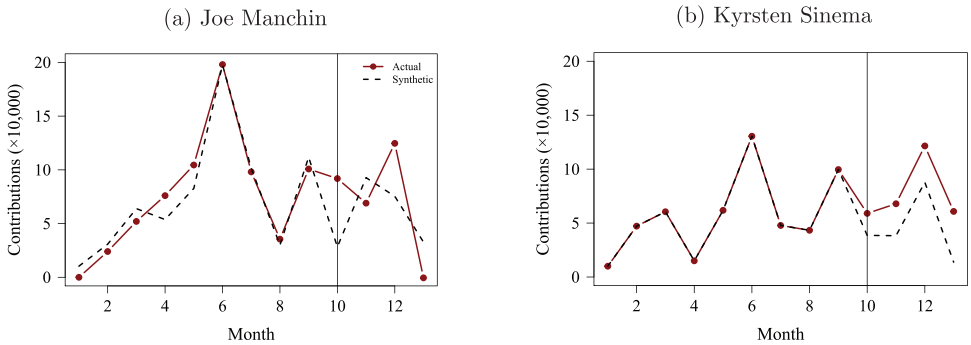

FIGURE 5 Synthetic control: Joe Manchin and Kyrsten Sinema. *Notes*: This figure shows the actual and synthetic contribution paths for Joe Manchin (a) and Kyrsten Sinema (b).

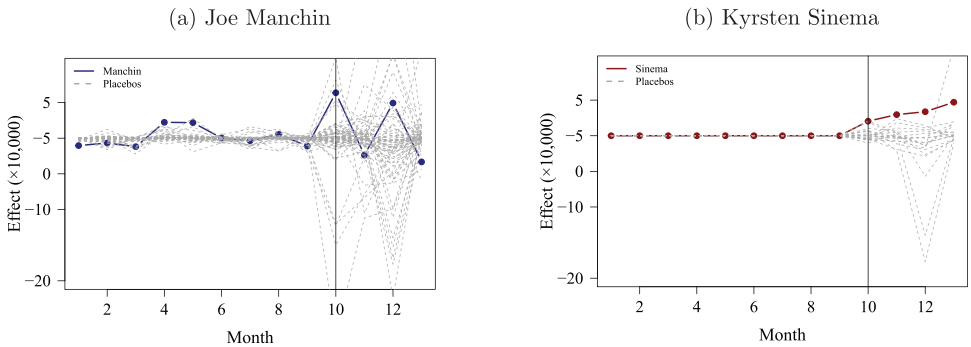

FIGURE 6 PreRMSE truncated placebos. *Notes*: Panel (a) plots the difference between the synthetic and control lines for Joe Manchin shown in blue, and the 70 other Senators who have a PreRMSE less than that of Manchin. Panel (b) plots the difference between the synthetic and control lines for Kyrsten Sinema shown in red, and the 19 other Senators who have a PreRMSE within 20 times that of Sinema.

squared error ratio (RMSER) for j:

$$\text{RMSER}_j = \frac{PostRMSE_j}{PreRMSE_j}.$$

Thus, a poor pretreatment fit is penalized by increasing the denominator of the RMSER, and a large estimated effect is rewarded by increasing the numerator. The convention has been to conduct a permutation test whereby the null hypothesis is that all of the RMSERs come from the same distribution. Then the p-value is the fraction of observed RMSERs that are greater than or equal to the RMSER of the treated unit. Because the theory specifically predicts an *increase* in contributions, I also report the one-tailed p-value that accounts for the sign of the treatment effect. The treatment effects, RMSERs, and p-values are all reported in Table 1, where it is again clear that Sinema received significantly more PAC contributions than she would have if BBB had not been on the agenda.[18]

TABLE 1 The effect of BBB on Manchin and Sinema's PAC contributions

	Joe Manchin	Kyrsten Sinema
Treatment effect	55,935.61	130,704.10
RMSER	3.58	2,257,094.00
Two-tailed p-value	0.66	0.04
One-tailed p-value	0.29	0.02

Note: This table includes the estimated treatment effects of the introduction of BBB on Manchin and Sinema's campaign contributions, along with the RMSER, and p-values. Abbreviations: BBB, Build Back Better Act; PAC, political action committee; RMSER, root mean squared error ratio.

One of the advantages of the SCM is that it yields weights that can be substantively interpreted, allowing researchers to assess whether the constructed synthetic unit aligns with domain knowledge. Table 2 presents the top 10 weights used to construct the synthetic control for Kyrsten Sinema, along with the corresponding senators. These weights are substantively plausible and reflect Sinema's distinctive

[18] For context, Sinema has the third lowest PreRMSE of all of the senators at .02, whereas Manchin is ranked 72nd at 12,613.12. The exceptionally low PreRMSE for Sinema raises valid concerns about overfitting. Yet, multiple robustness checks, reported in Online Appendix D (pp. 16–19), suggest that overfitting is not driving the results.

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TABLE 2 Top 10 synthetic control weights (Sinema)

Name (Party)	Weights
Patty Murray (D)	.17
Mark Warner (D)	.14
Tim Kaine (D)	.14
Ben Sasse (R)	.14
Jacky Rosen (D)	.05
Jerry Moran (R)	.03
John Hoeven (R)	.03
Bill Hagerty (R)	.01
Ron Johnson (R)	.01
Tom Carper (D)	.01

Note: This table includes the names and political party of the senators who have the top 10 weights in the synthetic control for Kyrsten Sinema.

character. Some of the largest weights are assigned to moderate Democrats such as Mark Warner and Tim Kaine, both of whom, like Sinema, have cultivated centrist reputations and represent politically competitive states. Patty Murray and Jacky Rosen, though generally more liberal, share salient characteristics with Sinema, including party affiliation and gender. Rosen, notably, is also a first-term senator from the neighboring state of Nevada. The inclusion of several Republican senators—most prominently Ben Sasse, Jerry Moran, and John Hoeven—suggests that the synthetic control is capturing cross-partisan similarities in policy emphasis and legislative behavior. Given Sinema's "maverick" political persona, and her appeal to both corporate and bipartisan constituencies, these weights appear to track many dimensions of her profile.

It may seem odd that Manchin was the one that publicly announced he would not vote for BBB, yet it was Sinema who received extra campaign donations as a result of BBB being on the agenda. There are a number of reasons, however, that might explain this surprising finding. First, it could simply be the case that the SCM failed to pick up the effect on Manchin due to the poor pretreatment fit. More substantively, Manchin is a well-known politician with close relationships with Democratic party leaders, despite being the most conservative Democrat.[19] He served in the Senate starting in 2010, and before that he was the governor of West Virginia. Being such a well-known and experienced politician with close relationships with party leaders means that he may have less need for campaign funds. Furthermore, Manchin announced in 2023 that he would retire from the Senate at the end of his term. In terms of the model, there was significantly less uncertainty about Manchin's preferences, and it would have cost groups far more to pressure Manchin, given his relative political security.

[19] On May 31, 2024, Manchin announced he was leaving the party.

Kyrsten Sinema on the other hand was a first term senator, starting her term in 2019, and before that, served three terms in the US House of Representatives. In December 2022, she left the Democratic Party to serve as an Independent. Thus, as a new senator with weak ties to the national Democratic Party, Sinema would have been more reliant on campaign donations, and her preferences were more uncertain to party leaders. After polling in 2024 suggested that she would likely lose in a three-way race against a Democrat and a Republican, she too chose to not run for a subsequent Senate term.

Finally, organized interests who wanted BBB to fail only needed a single vote to ensure their preferred outcome. Thus, it would have been optimal for them to coordinate on a single individual. Given the reasons listed above, Sinema would have been the likely focal point. Therefore, the findings of a large effect for Sinema and a positive but statistically insignificant effect for Manchin are consistent with the predictions of the model.

DISCUSSION AND CONCLUSION

The model in this paper yields a number of insights into the strategic interactions of vote buying and agenda setting. First, when there is sufficient uncertainty, vote buying can occur even though no formal vote is held. That is, the interest group can make an offer to the legislator to vote against a proposal, and in favor of the status quo. In these cases, when the agenda setter then learns that the legislator will not vote for the proposal, they will decide not to hold a formal vote. Because of negative agenda control, the decision of the legislator to vote against the proposal will not be recorded as a roll call vote. The conditions under which this happens are relatively general and thus, studies that regress roll call–based measures on campaign contributions to study the influence of money on voting behavior will likely underestimate such effects. This explains why more micro-level studies of vote buying such as Stratmann (2002), Dwyer and Stratmann (2023), and Grier et al. (2023)—that look at multiple votes that occur on the same piece of legislation—find significant effects, where studies based on aggregate roll call measures such as Ansolabehere et al. (2003) find null effects.

The second main insight of the model is that vote buying is most likely when the agenda setter and the interest group are extreme, the legislator is moderate, and there is high uncertainty over the legislator's preferences. One prominent case in which these conditions appear to have held is the 2021 negotiations over the BBB. To test whether vote buying may have influenced the outcome of BBB, I use a synthetic control design to construct counterfactual campaign

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contribution paths for the swing voters in the Senate, Joe Manchin and Kyrsten Sinema. I find that Sinema received significantly more contributions than she would have, had BBB not been on the agenda. This case therefore provides evidence that vote buying could have significant effects on public policy and legislative voting behavior in ways that would not be picked up by conventional research designs.

There are a number of avenues for future research building off of this framework. In the model here, the interest group does not have "access" to the proposer. This is to highlight the effect of vote buying, but it would be interesting to see how the results extend when the interest group can influence the proposer as well. The model also only includes a single interest group. There could be nuances that arise due to competition among vote buyers, or potentially also competition for access to the proposer. Finally, asymmetry in the aligned and misaligned cases suggest new dynamic incentives when the legislature shifts between the two. A fully dynamic model with endogenous status quo could therefore yield new insights.

The question of whether and how money influences the way legislators vote has important implications for the representativeness of the political process. In order to answer this question, many researchers study the empirical associations between voting behavior and campaign contributions. The model in this paper shows that this can go wrong in two ways. First, when the interest group is misaligned with the agenda setter and the legislator, the agenda setter will propose a policy that is biased toward the interest group in order to lessen the effect of vote buying. Thus, the mere potential for the group to buy votes leads to proposals that reduce their need to follow through on this threat. Second, when the group successfully buys votes against a proposal, the agenda setter will exercise negative agenda control, leading to this vote not being recorded in the data. Therefore, in order to empirically study the way money influences legislators' voting behavior, researchers need to take these strategic incentives into account, otherwise they are at risk of systematically underestimating the degree of interest group influence over both policy outcomes and legislators' voting behavior.

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